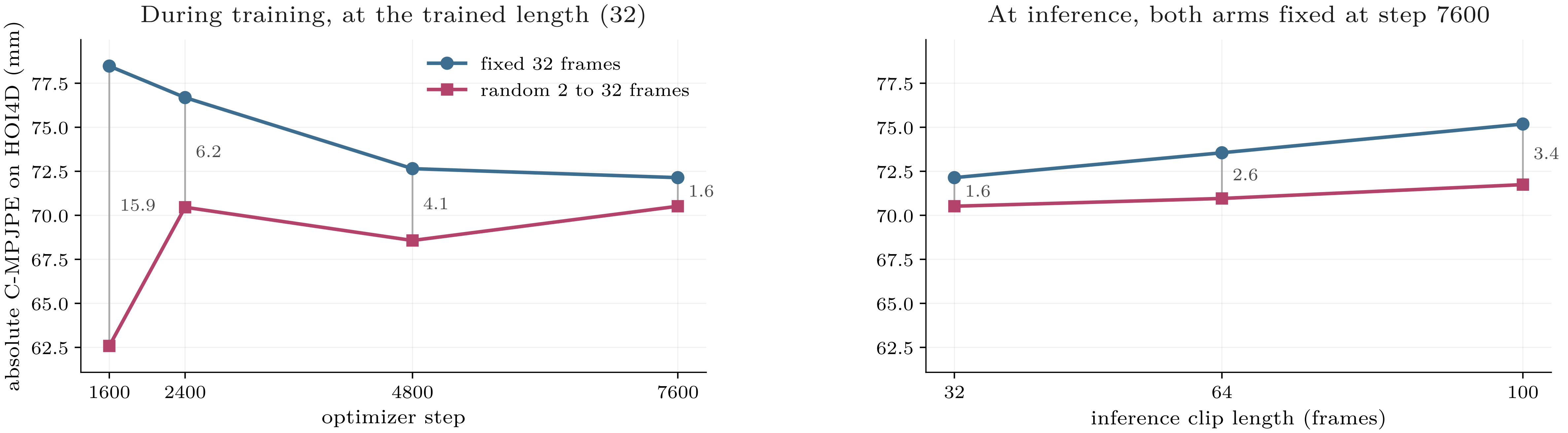
Random clip length buys length-robustness, not accuracy

Two identical runs, ARCTIC + OakInk2, HOI4D held out and scored zero-shot. Absolute C-MPJPE in mm, lower is better.



The gap COLLAPSES with training: 15.9 mm at step 1600, 1.6 at step 7600, both arms in epoch 10. Our seed band is ±0.8 mm on one seed per arm, so 1.6 mm at the trained length is not an effect we would defend.

The gap REOPENS with length: 32 to 100 frames costs the fixed arm +3.0 mm and the random arm +1.2. Neither was trained beyond 32, and the one that saw many lengths is 2.5× less sensitive to a length it never saw.

Recommendation

Adopt random clip length and claim length-robustness, not accuracy. It is free, it never loses, and it lets one trained model serve a window length chosen at test time. Do not quote the training-length gap as a result.

Both panels share the step-7600 checkpoints: the left panel's last point and the right panel's first point are the same two numbers.